



University of Frontier Technology, Bangladesh
Department of Internet of Things and Robotics Engineering

Course Code	CC 484
Course Title	Cloud Computing Sessional

LAB REPORT

On
Lab Final Task

Submitted To

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1. Objectives

1. The primary objective of this lab session is to gain practical experience with core Microsoft Azure services, including:
2. Deploying and configuring a Linux Virtual Machine (IaaS).
3. Managing unstructured data storage using Azure Blob Storage Account (PaaS).

Deploying a web application using Azure App Service (PaaS).

2. Task 1: Create and Configure a Linux Virtual Machine

2.1 Overview & Implementation Steps

4. Navigated to the Azure Portal and created a new Linux Virtual Machine under the Resource Group Lab1_RG.
5. Configured the VM name as LinuxVM1 in the Korea Central region running Linux (ubuntu 24.04) OS.
6. Selected the Standard D2s v3 size (2 vCPUs, 8 GB memory).

Configured network interfaces and successfully deployed the resource.

2.2 Deployment Verification & Network Properties

Deployment Status: Deployment complete successfully.

Resource Group: Lab1_RG

VM Name: LinuxVM1

Operating System: Linux (ubuntu 24.04)

VM Size: Standard D2s v3 (2 vCPUs, 8 GB memory)

Status: Running

Public IP Address: 40.82.128.203

Private IP Address: 172.16.0.4

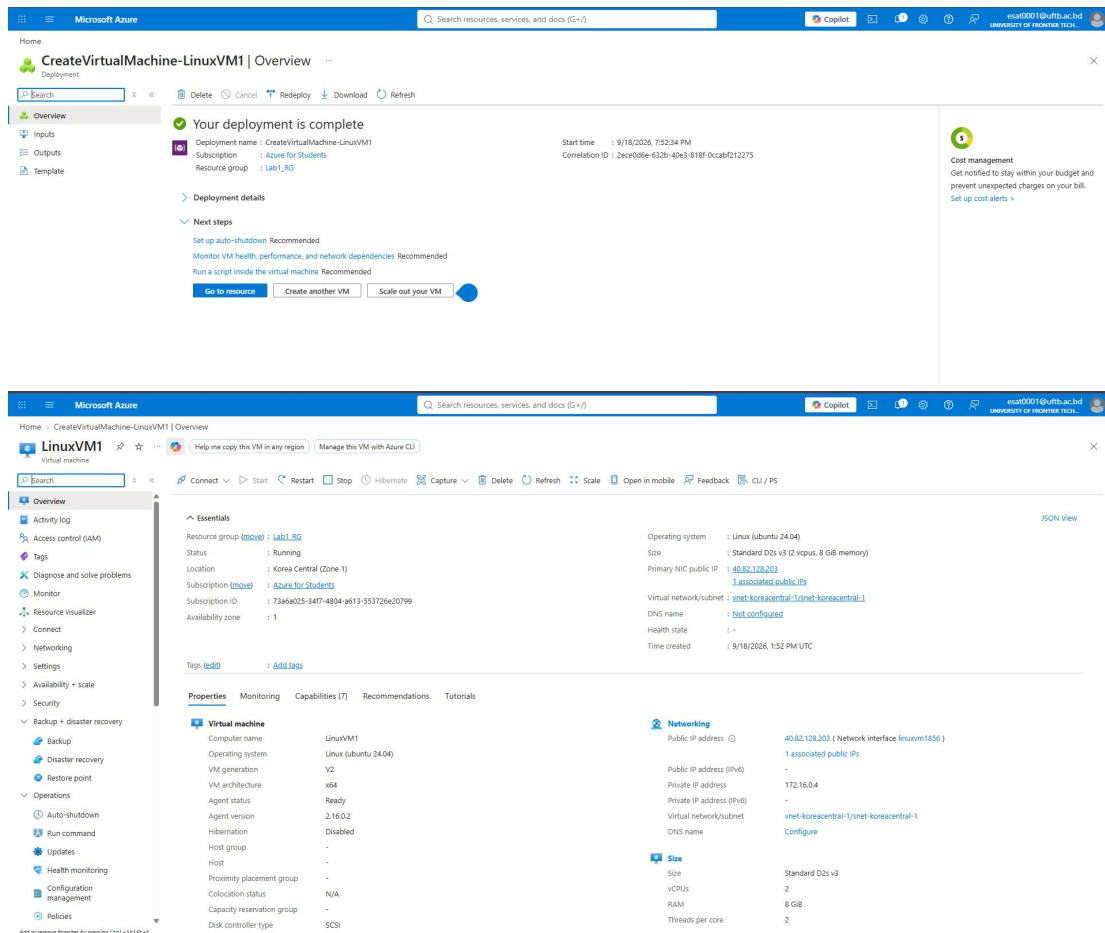


fig.Task 1: Create a Linux Virtual Machine

Task 2: Create Azure Storage Account and Blob Container

3.1 Overview & Implementation Steps

Created an Azure Storage Account named esatstorage2201022 under Lab1_RG in Korea Central.

Provisioned a Blob Container named sample-container.

Uploaded a file named sample.txt containing string data.

Verified direct access to the stored file via its generated URL (https://esatstorage2201022.blob.core.windows.net/sample-container/sample.txt[https://esatstorage2201022.blob.core.windows.net/sample-container/sample.txt]).

3.2 Key Technical Details

Storage Account Name: esatstorage2201022

Account Kind: StorageV2 (general purpose v2)

Replication: Locally-redundant storage (LRS)

Container Name: sample-container

Uploaded File: sample.txt (Size: 42 B, Content-Type: text/plain)

File Content: "Hello, Azure Storage!" This is Esat :))!

3.3 Theoretical Concepts

Storage Account Purpose: Serves as a unique, secure, and centralized namespace in Azure to manage scalable storage services (Blobs, Files, Queues, Tables).

Blob Container Purpose: Functions like a directory to organize unstructured binary large objects (files, images, media, text) accessible securely over HTTP/HTTPS.

The screenshot displays the Microsoft Azure portal interface for the storage account 'esatstorage2201022'. The left sidebar shows navigation options like Overview, Activity log, Tags, and various storage services. The main content area is divided into several sections:

- Essentials:** Provides basic information such as Resource group (koreacentral), Location (Azure for Students), Subscription ID (73ada025-34f7-4804-a513-553726e20799), and Disk state (Available).
- Properties:** Details the configuration for Blob and File services. For Blob service, settings like Hierarchical namespace, Default access tier, Blob anonymous access, Blob soft delete, Container soft delete, Versioning, Change feed, NFS v3, Allow cross-tenant replication, and Storage tasks assignments are listed with their current states (e.g., Disabled, Enabled, Not configured).
- Security:** Shows security-related settings, including Require secure transfer for REST API operations (Enabled), Storage account key access (Enabled), Minimum TLS version (Version 1.2), and Infrastructure encryption (Disabled).
- Networking:** Displays networking settings such as Public network access (Enabled), Public network access scope (Enable from all networks), Private endpoint connections (0), Network routing (Microsoft network routing), and Endpoint type (Standard).

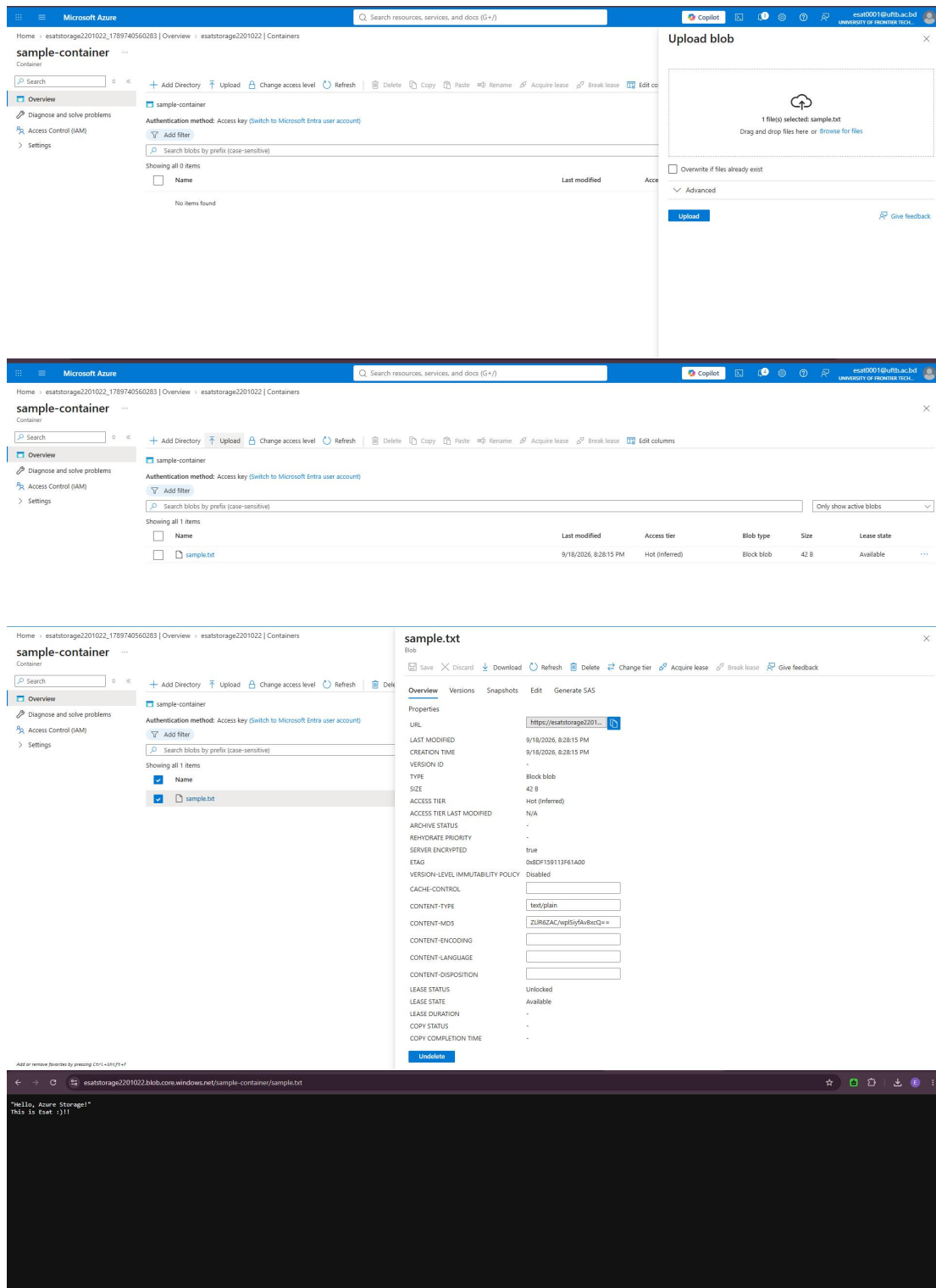


fig. Task 2: Create an Azure Storage Account and Blob Container

4. Task 3: Create Azure App Service Web App and Deploy Application

4.1 Overview & Implementation Steps

Created an Azure App Service instance named studentweb-2201022 targeting Php - 8.2 runtime stack on Linux OS.

Accessed the container management interface via **Advanced Tools (Kudu)** and connected using **SSH Application Container**.

Executed shell commands (cat << 'EOF' > ...) to deploy custom web application code directly to /home/site/wwwroot/index.php.

Formatted the application with a responsive CSS dashboard interface displaying student information.

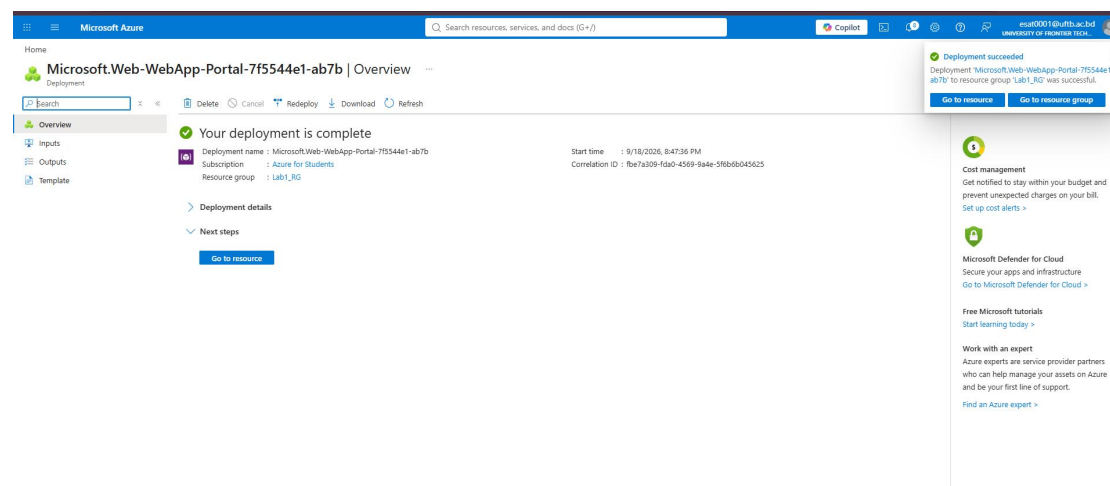
4.2 Application Details & Verification

App Name: studentweb-2201022

App Service Plan: ASP-Lab1RG-b069 (Basic SKU)

Generated URL: https://studentweb-2201022-cnh7dvhkf8c8atft.koreacentral-01.azurewebsites.net

Deployed Webpage Output: Displays "Welcome to Cloud Computing Lab", Student Names ("Mst. Esat Jahan Akhi & Sukria Akter"), and Student IDs ("2201022 & 2201021") inside a styled dark-themed UI container.



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Azure App Service

Overview

Dashboard

Monitoring

Tools

Diagnostic Tools

Configuration

studentweb-2301022 20240818.3 Instance: **761a992c6cbeec** [ip Switch](#)

SKU

Basic

OS

Linux

Logs & Diagnostics

Application

Platform

Log Stream

Download All Container Logs

Browse Log Files

Tools

SSH — Application

SSH — Kudu

File Manager

Environment

Settings & Variables

Deployments

List of Deployments

Last Deployment

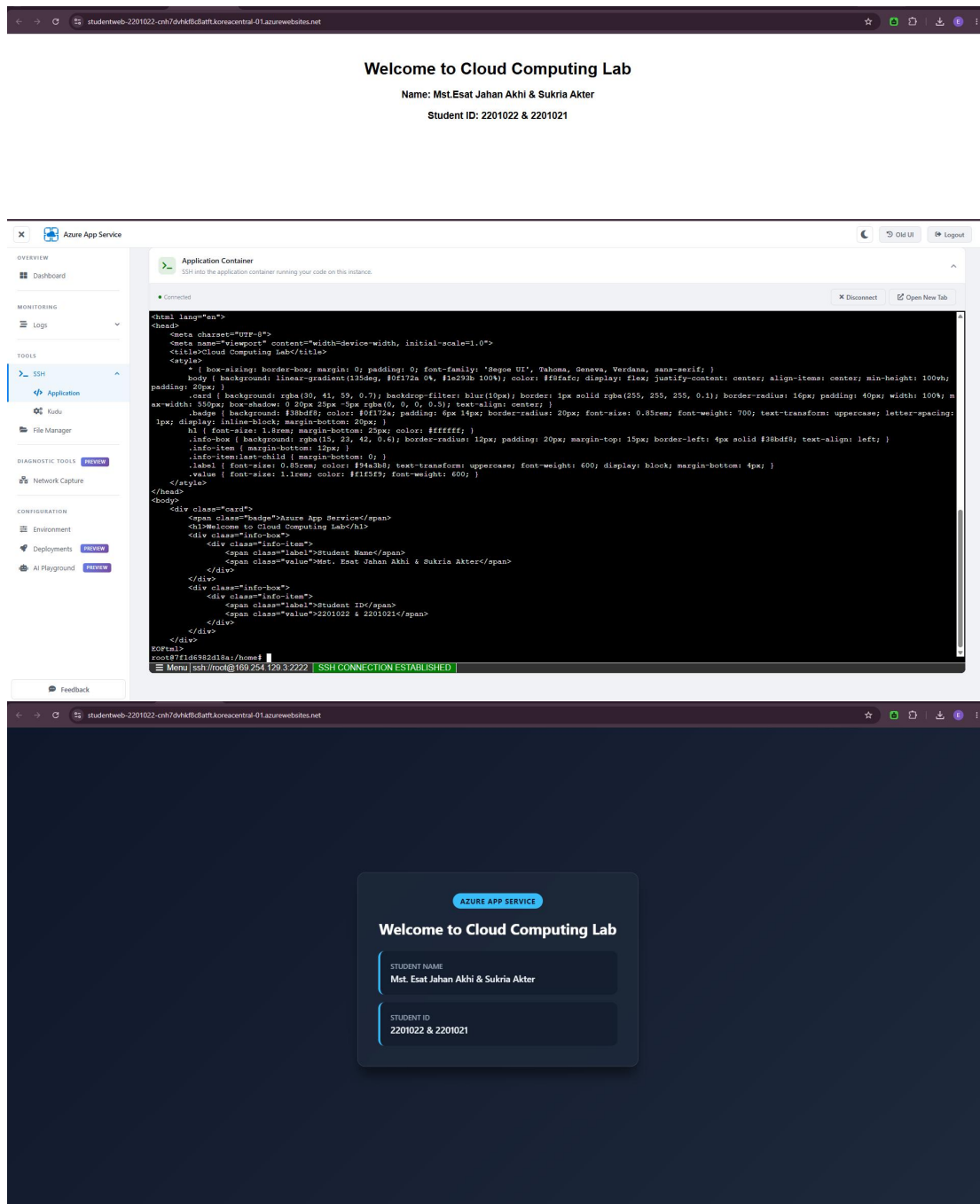


fig.Task 3: Create an Azure App Service Web App

5. Conclusion

All three tasks outlined in the lab guidelines were successfully executed. Virtual Machine infrastructure was instantiated, persistent object storage was configured with public file retrieval, and a live web application was deployed and served publicly via Azure App Service.